

Aaron Stone — Planet Fitness Routine

Lifter: Aaron Stone · **Gym:** Planet Fitness (full free-weight rack area + machine-rich) · **Split:** 4-Day Upper/Lower (Upper A / Lower A / Upper B / Lower B), each muscle ~2x/week · **Goal:** dual strength + physique, with a *therapeutic mandate* to train the upper body generously inside a stable shoulder ROM.

Bilateral-shoulder constraint ([exercise.constraint.aaron.shoulder-instability](#) — prior dislocation, rotator-cuff/labrum involvement, LIMITED external-rotation ROM, no pain): train WITHIN available ROM. **Programming rules in force on every page below:** neutral-grip / front-of-body pressing only (RULE 3); the apprehension position is BANNED — never load abduction + end-range external rotation, so no behind-the-neck press/pulldown, no deep dips, no forced wide-grip end-range stretch (RULE 4/5); controlled non-ballistic tempo (RULE 6); a programmed rotator-cuff + scapular-stability block TWICE weekly at 12 sets/week (RULE 1/RULE 2); intensity techniques kept OFF shoulder-loaded lifts (RULE 7); posterior-girdle volume held $\geq$ pressing volume (RULE 8); and a red-flag kill switch (RULE 9). The protection IS the training — do NOT under-dose the shoulder; sedentariness flares it.

Two gyms, two docs. This is Aaron's **Planet Fitness** routine. A separate document covers his **290 Revolution** home gym. Versus the 290 build, this PF gym adds machines the home setup lacks — a **leg press, pec-deck/machine chest press, machine lateral raise, hip-adduction machine, and standing + seated calf machines** — so the accessory volume here rides high-SFR machine paths around the real barbell rack work, rather than rigging everything on a Smith machine.

Generated from the [exercise](#) corpus against the Planet Fitness equipment menu and Aaron's constraint profile. **Not medical advice** — the rotator-cuff/scapular work is supportive prehab; keep any physio plan, and stop on any red flag.

How to read this / autoregulation

RIR targets (evidence-based). *Compound strength anchors* (the one heavier 5–8 rep neutral-grip press per upper day, the squat, the RDL/hinge) sit **3–4 RIR early-block, intensifying to 2–3 RIR late-block** — failure on big lifts is slightly inferior for strength and roughly *triples* acute neuromuscular fatigue vs ~3 RIR ([exercise.claim.17](#)), so 0–1 RIR is reserved for an end-of-block top set on safe patterns only. *Primary hypertrophy compounds* (rows, secondary presses, leg press, hinge accessories) sit **2–3 RIR, last set ~1**. *Isolation* (lateral/rear delts, curls, extensions, calves, leg curl, leg extension) runs **1–2 RIR with the final set of each exercise to 0–1 RIR** — a cheap place to spend the closer-to-failure premium ([exercise.claim.18](#)); any **light-load, high-rep** isolation set must reach **0–1 RIR** because motor-unit recruitment is recruitment-late at low loads.

Progression — double-progression + RIR autoregulation across a 4–5 week accumulation block, then deload. Add reps within each exercise's prescribed band at target RIR; once you hit the top of the band on all sets, add the smallest load increment and drop back to the bottom. **Dumbbells cap at 80 lb at this gym** — once a DB lift maxes out, progress via reps / added sets / tempo, never grip-width creep. Ramp volume from MEV (~8–12 sets/muscle) up through the MAV zone toward MRV across the block, biasing the *therapeutic* targets (lateral delt, rear delt, back, cuff/scap) to the generous end; ramp RIR **down** from ~4 to ~1 across the block (Martikainen — equal gains at lower average session-RPE); then deload to ~MV and restart at MEV.

Set-structure logic (from the corpus's central refutation). NO advanced technique out-grows volume-equated **straight sets** ([exercise.claim.13](#)), so straight sets anchor every shoulder-stakes compound (lowest fatigue, best load/technique control) and carry ~70% of weekly sets. Time-efficient structures are deployed only where they buy

efficiency without risking the joint: **antagonist supersets** on non-competing upper pairs and the leg-curl/leg-extension pair (Burke 2025 RCT — equal growth, ~36% less session time); **myo-reps** and a single capped **drop-set** on machine/cable isolation only; **cluster sets** optional on the heavy neutral-grip press for a velocity-maintained top-end strength bias.

Autoregulation — ordered so upper-body therapeutic volume is the LAST thing cut.

1. **RED-FLAG KILL SWITCH (overrides everything; RULE 9):** STOP the set and reassess on any of — a dislocation/subluxation event, sharp pain, a sense of the shoulder *slipping*/instability, numbness or tingling down the arm, or sudden loss of strength. These are joint-line/neural signals, categorically different from the dull working-muscle pump that is expected and fine. A red flag ends that exercise for the day; it does **not** mean cut the week's upper-body volume.
2. **Shoulder feels TIGHT (no red flag) → MODIFY the movement, do NOT remove volume.** In order: (a) shrink ROM to the stable/pain-free portion and keep the set; (b) swap to a friendlier pattern in the same target (neutral-grip / machine / cable instead of free-weight); (c) slow tempo and drop load, keeping 2–3 RIR; (d) front-load that day's cuff/scap block + face-pulls as a feeder to warm and stabilize the joint, then re-attempt the press. Only after these fail do you trim that single exercise's sets — and you **bank the removed sets into cuff/scap or rear-delt work** (RULE 8).
3. **Recovery poor (bad sleep/stress, can't match last session = MRV signal) → cut effort before volume, and cut the right volume first.** Raise RIR ~1–2 across the session; if still flat, trim sets from the *highest-fatigue* work first (heavy lower-body anchors, heaviest pressing) and drop the optional intensity techniques. The cuff/scap block, lateral/rear-delt, and back volume are the **protected last** things touched.
4. **Whole-muscle stall for 2+ sessions = that muscle hit MRV:** hold or shave 2–3 sets from THAT muscle only (not a blanket cut), or deload if several flag together.

Weekly schedule

Day	Session	Primary targets	Anchor (heavy 5–8)
Upper A	Horizontal-emphasis (push-bias) + Shoulder-Health Block #1	chest, lats/upper-back, lateral + rear delt, biceps, triceps, rotator-cuff/scapular	Dumbbell Flat Bench Press (neutral/semi-neutral grip)
Lower A	Knee-dominant (quad-emphasis)	quads (+ RF), glutes, hamstrings (hinge), adductors, gastrocnemius, core	High-Bar Back Squat
Upper B	Vertical-emphasis (front-of-body) + Shoulder-Health Block #2	front + lateral + rear delt, upper chest, lats, biceps, triceps, rotator-cuff/scapular	Machine Shoulder Press (fixed-path, front-of-body)
Lower B	Hip-dominant (hamstring/glute-emphasis)	hamstrings (both head groups), glutes, secondary quads, adductors, soleus, core	Romanian Deadlift (barbell)

A typical week runs Upper A / Lower A / rest / Upper B / Lower B / rest / rest, but any 4-day arrangement that keeps an upper and a lower not back-to-back works (frequency is a logistics lever, not an independent growth driver, once weekly volume is matched).

Weekly volume (therapeutic upper-body bias; cuff/scap a programmed priority)

Muscle	Weekly sets	Note
Chest (pectoralis major)	14	2×/wk. Upper A flat DB press 4 (heavy anchor) + machine press 3; Upper B incline Smith press 3 + indirect overhead/incline. All neutral/front-of-body, fixed-but-pain-free ROM (RULE 3/5).
Back / lats (lats, teres major, rhomboids, mid/lower trap)	18	2×/wk, a generous therapeutic ceiling (RULE 1). Upper A chest-supported incline row 4 + face-pull share; Upper B neutral-grip to-chest pulldown 4 + face-pull share. Neutral-grip, front-of-body, never behind-neck (RULE 4).
Anterior deltoid	10	Mostly indirect from pressing; the Upper B machine overhead press is the direct front-of-body driver (no barbell OHP, no behind-neck).
Lateral deltoid	18	2×/wk therapeutic priority — do NOT under-dose (RULE 1). Machine lateral raise both upper days (grip-free, core-free, controlled abduction arc).
Posterior (rear) deltoid	16	Kept ≥ pressing volume (RULE 8). Cable face pull both upper days + the rear-delt share of all rowing.
Rotator cuff + scapular stability	12 (a programmed priority, NOT an afterthought)	RULE 2: ~6 sets/upper day = cable external rotation (elbow-at-side, within available ROM) + serratus-punch scapular protraction + the face-pull's posterior-cuff share. Pinned at full dose from week 1; the <i>last</i> thing cut in a deload.
Biceps + brachialis	14	2×/wk EZ-bar curl, elbow-only, load in front of body — fully clear of the constraint.
Triceps	14 direct (+ indirect pressing toward the >20 total)	2×/wk rope pushdown, elbow-at-side; banks indirect volume from all pressing — triceps specifically reward >20 <i>total</i> weekly sets (Baz-Valle 2022).
Forearms (flexors/extensors/brachioradialis)	8	Mostly indirect from curls, rows, pulldowns, and the hinge grip.
Quadriceps	15	2×/wk. Lower A squat 4 + leg press 3 + reclined leg extension 3 (rectus-femoris bias); Lower B leg press secondary.
Hamstrings	14	2×/wk via BOTH patterns: hip-hinge (Lower A RDL + Lower B heavy RDL anchor — hip-extensor heads, long length) AND knee-flexion (Lower B seated leg curl).
Glutes (max + medius/minimus)	14	2×/wk, complementary lengths: long-length via squat + RDL, short/peak-length via the Lower B barbell hip thrust.
Adductors	8	2×/wk on the hip-adduction machine (a PF machine the 290 lacks) loaded into a deep abducted stretch; squat/leg-press hit them indirectly.
Calves (gastrocnemius + soleus)	12	Head split by knee angle: Lower A standing (knee-extended → gastroc), Lower B seated (knee-flexed → soleus).
Core (rectus abdominis, obliques, erector/QL)	10	~½ each lower day: Lower A hanging knee raise (lower rectus), Lower B ab-wheel rollout (anti-extension).

The 4 sessions

Upper A — Horizontal-Emphasis (Push-bias) + Programmed Shoulder-Health Block #1

Neutral-grip / front-of-body horizontal pressing and rowing, lateral + rear delts, direct arms, and the FIRST of two weekly rotator-cuff + scapular-stability blocks. One heavier 5–8 neutral-grip press is the strength anchor; everything else 8–15. Posterior-girdle (row + rear delt + cuff) volume is held $\geq$ pressing volume, and the cuff/scap work is a programmed priority. All pressing stays neutral-grip, front-of-body, controlled tempo, full-but-pain-free ROM — no apprehension position, no deep dips, no behind-the-neck, no forced end-range stretch.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
1	Dumbbell Flat Bench Press (neutral/semi-neutral grip)	4 × 6–8	3–4 (heavy press never below 2 RIR fatigued)	DB 65–75s (heavy end of the 80 lb cap; pick a pair leaving 3–4 RIR, ride double-progression toward the 80s)	straight-sets (run plain & first while fresh)	THE day's strength anchor in a shoulder-safe pattern: dumbbells let the wrists rotate to a neutral/semi-neutral grip in front of the body (RULE 3), stopping at a full-but-pain-free depth rather than a forced wide-bar end-range stretch (RULE 5); controlled ~2 s eccentric, no bounce (RULE 6).
2	Incline-Bench Chest-Supported Dumbbell Row (neutral grip)	4 × 10–12	2–3	DB 55–70s (chest-supported — grip/low-back not the limiter)	antagonist-superset (<i>pair with #3</i>)	Chest-supported removes lower-back and stabilization demand; a neutral/low-incline grip keeps the elbow in front of the torso plane and never drives it far behind the body under load (RULE 4). Posterior-girdle pull volume kept $\geq$ pressing (RULE 8).
3	Machine Chest Press (front-of-body, fixed path)	3 × 10–15	1–2 (last set 0–1)	machine ~mid stack (1–2 RIR at 12–15)	antagonist-superset (<i>pair with #2</i>)	Fixed-path, front-of-body horizontal press in a stable groove (RULE 3); stop the negative short of any unstable end-range stretch, keep the path smooth and non-ballistic (RULE 5/6). High SFR adds pec volume without extra joint stress after the heavy free-weight work.
4	Machine Lateral Raise	4 × 12–20	0–1 on the final set	machine ~light-to-mid stack	antagonist-superset (<i>pair with #5</i>); myo-reps optional on the last set later in block	Pure frontal-plane abduction, no external-rotation/apprehension component; grip/core-free path lets Aaron stop the moment the joint feels unstable (full-but-pain-free ROM). Lateral-delt volume is therapeutic and stays generous (RULE 1).
5	Cable Face Pull	3 × 15–20	1–2 (controlled, never grinding)	cable ~light-to-mid (lead with elbows, pull to forehead height, ER only within available ROM)	antagonist-superset (<i>pair with #4</i>)	Double-duty rear-delt + posterior-cuff work and a core piece of the programmed cuff/scap block (RULE 2). Keep ER strictly within available ROM; do NOT let the elbows travel into the abducted end-range-ER apprehension position (RULE 4); controlled tempo (RULE 6).

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
6	Cable External Rotation (elbow at side)	3 × 12– 20	1–3 (low-fatigue, never to failure)	cable ~lightest usable (continuous tension through the pain-free arc; elbow pinned to side)	straight-sets	Direct rotator-cuff strengthening at the neutral elbow-by-side angle — the safe, NON-apprehension position (RULE 4): no abduction + end-range ER. The therapeutic centerpiece — the protection IS the training (RULE 1/2). Stop at the first sign of instability; train only the pain-free range.
7	Supine Cable Scapular Protraction (serratus punch)	3 × 12– 20	1–3	cable ~light (or a single light DB pressed straight up, protracting the scapula at the top)	straight-sets	Completes the scapular-stability half of the cuff/scap block (RULE 2): trains scapular protraction / upward-rotation control with minimal glenohumeral motion, building the serratus support that stabilizes the shoulder for pressing. Low fatigue, controlled, never near failure.
8	EZ-Bar Curl	3 × 8–12	1–2 (last set 0–1)	EZ-bar ~50–70 lb (double-progress within the band)	antagonist-superset (<i>pair with</i> #9)	Direct elbow-flexion with the load in front of the body — no shoulder ER/abduction demand, fully clear of the constraint. EZ grip is wrist-friendly.
9	Rope Triceps Pushdown	3 × 12– 20	0–1 on the final set	cable ~mid stack (spread the rope at the bottom for end-range lockout)	antagonist-superset (<i>pair with</i> #8); one capped drop-set allowed later in block	Elbow-at-side pushdown keeps the shoulder neutral and unloaded — no overhead or behind-the-body position. Banks direct triceps on top of indirect pressing from #1–3.

Banked this day vs the ~2×/week weekly targets: chest ~7 (toward 14) · back/lats + upper-back ~7 (toward 18) · lateral delt ~4 (of 18) · rear delt ~7 (half of 16) · rotator-cuff/scapular ~6 = **exactly half the 12/wk therapeutic dose** · biceps ~4 and triceps ~4 + indirect pressing (toward 14 each).

Structure & rest. Run three antagonist supersets to preserve volume in less time (Burke 2025, ~36% time saved) — A2+B (press/row), E+D (face-pull/lateral-raise), G+H (curl/pushdown) — but treat the heavy DB press (#1) as plain straight sets first while fresh. Rest: ~3 min on the heavy DB press · ~90–120 s shared on each antagonist superset · ~60–90 s on isolation · ~45–75 s on the cuff/scap work (never to grinding failure). Loads are week-1 entry references at the higher-RIR end of the descending block ramp.

Shoulder call-out (Upper A). KILL SWITCH (RULE 9): stop and reassess on any subluxation/slip sensation, sharp joint-line pain, arm numbness/tingling, or sudden strength loss — a dull muscle pump is fine. If the shoulder feels *tight* (no red flag): shrink ROM to the stable portion, slow tempo, or front-load the external-rotation + serratus work as a feeder, then re-attempt the press — **modify the movement, do NOT cut the upper-body volume**. The cuff/scap block (orders 5–7) is the literal mechanism that resolves Aaron's symptoms — it runs at full dose from week 1 and is the last thing ever trimmed.

Cites: [exercise.claim.13-advanced-techniques-class-vs-volume-equated-straight-sets](#) , [exercise.claim.mu-flat-press-mid-pec-bias](#) , [exercise.tech.antagonist-superset](#) (Burke 2025 RCT), [exercise.claim.mu-row-lat-tension-vs-pulldown-leverage](#) , [exercise.claim.18-zero-rir-beats-low-rir-closer-is-more-growth](#) , [weekly_sets](#) (triceps >20 total, Baz-Valle 2022), [shoulder_accommodations](#) RULE 1/2/4/8, [exercise.constraint.aaron.shoulder-instability](#) .

Lower A — Knee-Dominant (Quad-Emphasis)

Squat-pattern knee-dominant strength anchor (heavier 5–8), then quad-biased leg press + rectus-femoris-biased leg extension, one long-length hamstring/glute hinge (RDL), an adductor machine, a standing knee-extended calf raise (gastrocnemius), and a hip-flexed core finisher. A real barbell back squat in the rack anchors the day for Aaron's functional-strength goal; PF's leg press, leg extension, hip-adduction, and calf machines supply high-SFR accessory volume. Carries ~half the weekly quad / hamstring / glute / adductor / calf / core targets.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
1	High-Bar Back Squat	4 × 5–8	3–4 early- block, 2–3 late- block	barbell ~155–185 lb to start (Olympic bar, no effective ceiling here); +10 lb when all sets hit 8 at target RIR	straight-sets	Day's heavy anchor on a REAL barbell for functional strength; bar racked across the traps front-of-body, never overhead/behind-neck and never the apprehension position (RULE 3/4) — shoulder-safe. Sit straight down through deep knee-flexion ROM to keep quads loaded at long length. If the rack position bothers the cuff, sub Front Squat or Smith Machine Squat (same quad target, fixed path).
2	Leg Press	3 × 10–15	2 (last set ~1)	machine ~3–4 plates/side (~270–360 lb on the sled); low/narrow foot placement to bias the quads	straight-sets	High-SFR vastii-dominant quad volume on a PF machine the 290 lacks — the spinal/systemic offload lets Aaron pile on hard quad sets after the squat. Low/narrow feet lengthen the quad's knee-flexion ROM for a quad bias.
3	Romanian Deadlift (RDL)	3 × 6–10	2–3	barbell ~135–185 lb to start; hinge to a deep but braced stretch	straight-sets	Long-length hip-hinge to cover this day's hip-extensor hamstring + glute + erector share on a real barbell (functional posterior-chain strength); the knee-flexor heads get their dedicated seated-curl share on Lower B.
4	Leg Extension (reclined / extended-hip, rectus-femoris bias)	3 × 12–20	0–1 (final set to failure)	machine ~mid-stack; recline the seat to ~40° hip flexion	straight-sets; optional single capped drop-set on the final set late-block	Hip-extended (reclined) seating loads the biarticular rectus femoris at long length — the one quad head the squat/leg-press under-bias — to genuinely hit EVERY quad head. Cheap place to chase 0–1 RIR (low systemic cost).
5	Hip Adduction Machine	3 × 12–20	0–1	machine ~lower-to-mid stack; open the pads to a deep abducted stretch for full ROM	straight-sets	Direct adductor volume on a PF machine the 290 lacks — covers ~half of the 8 weekly adductor sets (squat/leg-press only hit them indirectly). Deep abducted stretch loads them at long length.
6	Standing Calf Raise (machine)	4 × 10–20	0–1	machine ~mid-stack; pause and get a full bottom stretch each rep	straight-sets	STANDING / knee-EXTENDED on purpose: this is Lower A's gastrocnemius (long-length) session; the SEATED soleus work lives on Lower B so both heads get distinct, length-appropriate stimulus. Emphasize the lengthened bottom range.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
7	Hanging Knee Raise	3 × 15– 25	0–1	bodyweight (add a light DB between the feet once 25 reps is easy)	straight-sets	Lower-rectus-abdominis core finisher (~half the 10 weekly core sets); no shoulder loading in the apprehension position — a passive dead-hang grip with the arms in front of the body. If grip/shoulder fatigue, swap to the Cable Crunch (kneeling, fully shoulder-neutral).

Structure & rest. One heavy 5–8 knee-dominant anchor (squat) at 3–4 RIR early-block intensifying to 2–3 late-block; everything else 8–15 at the moderate-load sweet spot, isolation closer to failure. Rests: 3–4 min on the squat (up to 5 on top sets) · 2–3 min on leg press/RDL · 60–120 s on isolation. Run leg extension as straight sets (no leg-curl pair on this day). Standing (knee-extended) calf raise is deliberate so the gastroc gets its long-length stimulus here; the seated/soleus work lives on Lower B (Kinoshita).

Shoulder call-out (Lower A). The shoulder is essentially uninvolved on a lower day, but the back-squat bar is racked front-of-body across the traps (NOT overhead, NOT behind-neck) and stays out of the apprehension position. If the high-bar rack position pinches the cuff, swap to **Front Squat** or **Smith Machine Squat** — both honor RULE 3/RULE 4 and keep the same quad target.

Cites: [exercise.claim.ml-knee-dominant-squat-quad-bias](#), [exercise.claim.ml-deep-squat-more-leg-growth](#), [exercise.tech.straight-sets](#), [exercise.claim.ml-leg-press-foot-position-bias](#), [exercise.claim.ml-rdl-hip-dominant-vs-leg-curl-knee-dominant-hamstring](#), [exercise.claim.ml-leg-extension-extended-hip-rectus-femoris](#), [exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated](#), [exercise.claim.2-standing-vs-seated-calf-raise-gastroc](#) / [exercise.claim.ml-standing-vs-seated-calf-gastroc-soleus](#), [exercise.tech.drop-set](#).

Upper B — Vertical-Emphasis (front-of-body) + Programmed Shoulder-Health Block #2

Neutral / front-of-body OVERHEAD pressing (fixed-path machine, NO barbell behind-neck, NO ballistic) as the heavier 5–8 strength anchor; FRONT-of-body vertical pulling (neutral-grip to-chest pulldown); incline neutral press for upper-pec angle variety; a high-priority therapeutic lateral- + rear-delt repeat; arms; and the SECOND weekly programmed rotator-cuff + scapular-stability block. Every press is neutral/front-of-body and stays out of the apprehension position, controlled tempo, full-but-pain-free ROM. Posterior-girdle volume is kept ≥ pressing volume.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
1	Machine Shoulder Press (fixed-path, neutral/front-of-body overhead anchor)	4 × 6–8	3–4 (week 1; → 2–3 late-block, top set may touch 1 only on this safe fixed path)	machine ~mid- to-upper stack (start ~60–70% stack so 6–8 lands at 3–4 RIR; double-progress up). Fixed-path, NOT a barbell	straight-sets (double progression); cluster-set is the only alt if a top- end strength bias is wanted	Fixed-path machine keeps the press in a guided front-of-body groove — no behind-neck, no apprehension position, no ballistic drive; press only to a full-but-stable lockout, controlled ~2 s eccentric, stop short of any slip/pinch. This is intentionally the machine slot, not a barbell OHP.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
2	Close Neutral-Grip Lat Pulldown (front-to-chest, neutral grip)	4 × 10–12	2–3 (last set ~1)	machine/cable ~mid stack (a load failing ~12–14; leave 2–3 in the tank at 10–12)	antagonist-superset (<i>pair with #3</i>)	Neutral grip, pulled to the CHEST in front of the body — never behind the neck (RULE 4 bans behind-neck pulldown); wrist-friendly neutral handle, full but pain-free overhead reach without forcing end-range external rotation at the top.
3	Incline Smith Machine Press (fixed-path upper-pec, front-of-body)	3 × 8–12	2–3 (last set ~1)	smith ~95–115 lb to start (fixed bar path; ~30° bench for clavicular pec)	antagonist-superset (<i>pair with #2</i>)	Front-of-body fixed-path press at a moderate (~30°) incline — keeps the anterior delt out of the deep wide-grip end-range stretch and out of the apprehension position; press to a full-but-stable lockout, controlled tempo, no bounce. Stop the descent at the first sign of instability rather than forcing a deep stretch.
4	Machine Lateral Raise (high-priority therapeutic repeat)	4 × 12–20	1–2 (final set 0–1)	machine ~low-to-mid stack (grip-free, core-free)	straight-sets, optional single capped drop-set on the LAST set only	Lateral delt is a generous THERAPEUTIC target — do NOT under-dose it (RULE 1). Machine path raises within the safe scapular/abduction arc without grip or end-range ER; keep reps controlled, stop short of any apprehension at the top, full-but-pain-free ROM.
5	Cable Face Pull (rear delt + posterior cuff + scapular retraction)	3 × 15–20	1–2	cable ~low stack (light enough to keep ER WITHIN available ROM and controlled)	antagonist-superset (<i>pair with #4</i>)	Part of the day's cuff/scap block: pull to the face with elbows high but keep ER INSIDE Aaron's available ROM — do not drive the elbows behind the torso plane or into the apprehension position (RULE 2/4/8). Light load, controlled, never to grinding failure. Doubles as the band-pull-apart for scapular retraction.
6	Cable External Rotation (elbow at side) — programmed cuff work	3 × 12–20	1–3 (never to grinding failure)	cable ~lightest usable (continuous tension; elbow pinned to side, neutral abduction)	straight-sets — NOT a slot for any intensity technique (RULE 7)	Core of the therapeutic block — direct posterior-cuff ER at the elbow-by-side angle, only THROUGH Aaron's available ER ROM (do not force past neutral/end-range). Light, slow, pain-free; stop at the first hint of instability. This is the protection-IS-the-training work.
7	EZ-Bar Curl + Rope Triceps Pushdown (antagonist superset)	3 each, alternated	1–2 (final set 0–1)	EZ-bar ~50–70 lb; rope pushdown ~mid cable stack	antagonist-superset (biceps + triceps, non-competing)	Both are elbow-only movements with no shoulder-position demand — fully shoulder-safe; wrist-friendly EZ-bar and rope keep the joints comfortable. Triceps also bank indirect volume from the day's pressing.
8	Supine Cable Scapular Protraction (serratus punch) — scapular-stability closer	2 × 12–20	1–3	cable ~light (open-chain protraction; minimal glenohumeral motion)	straight-sets	Completes the scapular-stability half of the cuff/scap block (protraction to complement the face-pull's retraction). Minimal glenohumeral movement keeps load off the unstable joint while training the serratus that controls scapular upward-rotation for safe pressing. Controlled, pain-free.

Structure & rest. The overhead anchor is a FIXED-PATH machine press — the one slot deliberately kept OFF the barbell per the brief's overhead rule (barbell bench/squat/DL are fine for Aaron elsewhere). Face pull doubles as the

band-pull-apart (rear delt + infraspinatus/teres minor + mid-trap retraction); with the serratus punch and cable ER that is the day's ~6-set cuff/scapular share (12/week across both upper days). Rest: ~3 min on the overhead anchor · 2–3 min on the neutral pulldown/incline press · 60–120 s on isolation · 45–75 s on the cuff/scap work (paired antagonist work shares ~90–120 s). Loads are week-1 accumulation starting points.

Shoulder call-out (Upper B). The overhead anchor is intentionally a machine, not a barbell OHP — no bar-behind-neck, joint guided through a safe groove. ER work is loaded WITHIN available ROM only — stop short of any apprehension/slip, keep it light at 1–3 RIR, never grind. Lateral and rear delt are the generous therapeutic targets — do NOT trim them; if the shoulder feels tight, shrink ROM or swap to the machine/cable version and **KEEP the volume, banking any removed pressing sets into cuff/rear-delt work** (RULE 8). Red-flag kill switch applies (slip/subluxation, sharp pain, numbness/tingling, sudden strength loss → stop and reassess).

Cites: [exercise.tech.straight-sets](#), [exercise.claim.13](#), [exercise.claim.17-failure-required-to-maximize-growth-and-strength](#), [exercise.tech.antagonist-superset](#) (Burke 2025 RCT), [exercise.claim.19-heavy-load-required-for-hypertrophy](#), [exercise.claim.18](#), [exercise.tech.drop-set](#), [shoulder_accommodations](#) RULE 1/2/3/4/5/7/8, [exercise.constraint.aaron.shoulder-instability](#).

Lower B — Hip-Dominant (Hamstring/Glute-Emphasis)

Hip-hinge strength anchor (heavier 5–8 barbell RDL), then knee-flexor hamstring, peak-contraction glute, secondary quad, adductor, and the SOLEUS calf head (seated/knee-flexed — covering the head Lower A's standing work skips) at 8–15, plus an anti-extension core. Carries ~half the weekly hamstring/glute/adductor/calf volume. Shoulder-neutral day: all loading is front-of-body or machine, nothing touches the apprehension position.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
1	Romanian Deadlift (RDL)	4 × 5–8	3–4 (week 1) → ~2 late-block; reserve 0–1 for an end-of- block top set only	barbell ~185 lb (135 lb bar+plates to start, climb toward 225+)	straight-sets	Front-of-body double-overhand grip, controlled ~2 s eccentric, hinge to a full-but-stable hamstring stretch — never the apprehension position; this is the day's REAL barbell functional-strength anchor.
2	Seated Leg Curl	4 × 10–15	1–2, last set to 0–1	machine ~mid stack (~100–120 lb), progress reps-then-pin	antagonist-superset (<i>pair with #3</i>)	Flexed-hip seated position loads the hip-extensor hamstring heads at long length; superset back-to-back with the leg press, then ~90–120 s shared rest.
3	Leg Press	3 × 10–15	1–2	~3–4 plates/side (~315–405 lb incl. sled); high/wide foot placement to bias glute/ham on this hip day	antagonist-superset (<i>pair with #2</i>)	PF machine the home setup lacks — secondary quad volume + glute with zero spinal load, sparing the RDL-fatigued erectors; superset with the leg curl.

#	Exercise	Sets × Reps	RIR	Load	Set-structure	Notes / accommodation
4	Barbell Hip Thrust	3 × 10–15	1–2	barbell ~225 lb (start ~135–185, climb); pad the bar	straight-sets	Highest-EMG bilateral glute loader at the SHORT/peak-contracted length — complements the long-length stretch the RDL/squat give the glutes; spine-friendly, drive through heels to a hard lockout.
5	Hip Adduction Machine	3 × 12–20	0–1 (final set)	machine ~mid stack; open pads to a deep abducted stretch for full ROM	straight-sets	Exploits a PF machine the home gym lacks; high-SFR adductor volume (~half the weekly 8 sets) loaded into the lengthened position.
6	Seated Calf Raise (machine)	3 × 15–30	0–1	machine ~mid stack; full stretch at the bottom, pause at top	straight-sets; myo-reps optional when time-limited	Knee-FLEXED seated position targets the soleus — the head Lower A's standing/knee-extended work skips — so the gastroc/soleus split is covered across the week.
7	Ab-Wheel Rollout	3 × 8–15	1–2	bodyweight (shorten ROM to scale; kneeling start)	straight-sets	Anti-extension core to balance the day's heavy hinge loading on the erectors/QL; high-SFR, lengthened-range bias. Keep the rollout within a stable shoulder range (no overhead end-range stress).

Structure & rest. Straight-sets/double-progression on the RDL anchor, hip thrust, and leg press (the lowest-fatigue benchmark, ~70% of sets). Antagonist superset on seated leg curl + leg press (Burke 2025 RCT — same growth, ~36% less time; knee-flexor hamstrings rest while quads/glutes work). Myo-reps available on the seated calf finisher when time-limited. Hamstrings get BOTH a hip hinge (RDL — hip-extensor heads at long length) and a knee-flexion curl (seated — semitendinosus/distal BF). Calf head split: Lower A = standing/gastroc, Lower B = seated/soleus (Kinoshita). DB cap is 80 lb here, so the **barbell** RDL (not DB) is the correct heavy hinge anchor; erector-spinae fatigue from the RDL is the practical ceiling on the day, which is why the spine-offloaded hip thrust and leg press follow it rather than a second free-weight hinge. Rests: 3–4 min on the RDL, 2–3 min on leg press/hip thrust, 60–120 s on isolation.

Shoulder call-out (Lower B). A shoulder-neutral day — all loading is front-of-body or machine, nothing touches the apprehension position, so the joint just gets a low-stress hinge grip (no overhead / end-range ER anywhere). The RDL grip is double-overhand and front-of-body; the ab-wheel rollout is kept within a stable shoulder range.

Cites: [exercise.tech.straight-sets](#) , [exercise.claim.ml-rdl-hip-dominant-vs-leg-curl-knee-dominant-hamstring](#) , [exercise.claim.ml-seated-vs-lying-leg-curl-length-bias](#) , [exercise.tech.antagonist-superset](#) (Burke 2025 RCT), [exercise.claim.ml-leg-press-foot-position-bias](#) , [exercise.claim.ml-hip-thrust-short-length-squat-long-length-glute](#) , [exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated](#) , [exercise.claim.ml-standing-vs-seated-calf-gastroc-soleus](#) , [exercise.tech.myo-reps](#) .

Mesocycle (4–5 week accumulation block → deload, then restart at MEV)

Volume ramps MEV → MAV across the block, biased to the therapeutic upper targets; RIR ramps **down** from ~4 to ~1 (Martikainen — equal gains at lower average session-RPE). The **rotator-cuff + scapular block runs at its full 12 sets/week from week 1** — it is a therapeutic constant, never a ramp variable.

Week	Focus	RIR	Volume
1 — Accumulation entry / re-groove	Bed in bar path on the four anchors (DB flat press, squat, machine overhead, RDL) at heavy 5–8 but LIGHT effort; confirm the neutral/front press groove is pain-free through full available ROM. Straight-sets only this week so every load/RIR is a clean reference. Both cuff/scap blocks already at full dose.	Anchors ~4 (RPE 6); primary compounds 3; isolation 2	START at MEV, generous for therapeutic upper targets: lateral delt ~14, rear delt ~12, back ~14, chest ~11, biceps/triceps ~11 each, quads ~12, hams ~11, glutes ~11, calves ~10 — plus the fixed 12-set cuff/scap block (does NOT start low).
2 — First double-progress on step	Attempt the top of each rep band at week-1 load before adding the smallest increment (DBs cap at 80 lb). Anchors begin loading at the heavy 5–8 end. Antagonist supersets now standard (row+press, face-pull+lateral-raise, biceps+triceps, leg-curl+leg-press).	Anchors ~3–4; primary compounds 2–3; isolation 1–2, last set of each to 0–1	RAMP +1–3 sets into PRIORITY muscles only: lateral delt ~16, rear delt ~14, back ~16, triceps ~12; hold lower body and chest. Cuff/scap stays pinned at 12.
3 — Accumulation peak / push toward MAV	Most lifts now adding load or sitting mid-band. Bring lower body, chest, biceps up to the upper-middle of their range. Introduce time-efficient intensity structures ONLY on machine/cable isolation (myo-reps or one capped drop-set on lateral raise / leg extension / calf / pushdown) — kept OFF every shoulder-loaded lift (RULE 7). Optional clusters on the heavy machine press.	Anchors ~3; primary compounds ~2 (last set ~1); isolation 1, finals to 0–1	Near top of the MAV climb: lateral delt ~18, rear delt ~16, back ~18, chest ~13–14, quads ~14–15, hams ~13–14, glutes ~13–14, arms ~13–14, calves ~12. Cuff/scap still at 12.
4 — Intensification (effort over added volume)	Volume HOLDS at the week-3 ceiling; the lever shifts to LOAD on the anchors and proximity-to-failure on hypertrophy work. Genuine 5–8 strength zone expressed; optional reverse-pyramid on the LOWER-body anchors only (AVOIDED on overhead/unstable-shoulder pressing). Hold any muscle that flags an MRV signal.	Anchors ~2–3 (0–1 only on safe lower/neutral-press lifts); primary compounds ~1–2; isolation 0–1	HOLD at week-3 levels; redirect adaptation into intensity, not sets. If a muscle missed performance, drop IT ~2–3 sets only. The cuff/scap and lateral/rear-delt volume are the LAST things touched.
5 — Intensification peak (optional)	Run it only if recovery still tracks (else collapse to a 4-week block + deload). Top-end loads on the four anchors at lowest planned RIR; no new structures — shoulder-stakes lifts stay on plain straight-sets/clusters at controlled tempo (RULE 6). End-of-block top sets are the ONLY place 0–1 RIR is permitted on a heavy press, and only on the neutral/front patterns Aaron presses pain-free.	Anchors ~2 (a single top set may touch 0–1 on safe patterns only); primary compounds ~1; isolation 0–1 / finals to failure	Still HELD at the block ceiling — a brief touch of MRV to trigger supercompensation, not a new volume PR.

Week	Focus	RIR	Volume
Deload (week 5 or 6)	Fatigue flush, not a layoff (sedentariness flares the shoulder). RIR FIRST: raise effort to 3–6 RIR on everything. Then cut volume toward MV (~6 sets) on the high-fatigue work only — heavy lower-body anchors + heaviest pressing halved, loads –10–20%. CRITICAL ASYMMETRY: do NOT cut the upper body to MV uniformly — the cuff/scap block stays at/near its full 12 sets (low-fatigue, the literal mechanism that resolves the symptoms); lateral/rear-delt and back drop only modestly (~8–10 sets). After deload, restart the next mesocycle at MEV with RIR back at ~4.	3–6 RIR everywhere	High-fatigue compounds → ~MV (6 sets); cuff/scap held ~12; lateral/rear-delt/back ~8–10.

Mesocycle cites: [exercise.tech.straight-sets](#), [exercise.tech.antagonist-superset](#), [exercise.tech.myo-reps](#), [exercise.tech.drop-set](#), [exercise.tech.cluster-set](#), [exercise.tech.reverse-pyramid](#), [exercise.claim.17](#), [exercise.claim.18](#), [exercise.claim.21](#), [weekly_sets](#) (MEV/MAV/MRV/MV landmarks; triceps >20 total), [frequency](#) (2×/week floor), [shoulder_accommodations](#) RULE 1/2/6/7, [exercise.constraint.aaron.shoulder-instability](#).